SEA SHORE

BEACH HACK

JAN 30 31 & FEB 1

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PROBLEM DEFINITION

- Homes consume electricity blindly
- Users don't know which appliance increases the bill
- 90% households lack appliance-level awareness
- EVs, solar, inverters increase usage
- No real-time monitoring solutions available

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PROPOSED SOLUTION

- AmpAware smart energy monitoring system
- ESP32 + CT clamp hardware
- Android app with live energy data
- Smart alerts for abnormal usage
- Helps users monitor, plan, and control energyAdd a subheading

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MARKET ANALYSIS AND OPPORTUNITY

- TAM: 25 million Indian households
- SOM: 1 million Kerala households
- Initial target: 100,000 homes
- Market growing due to EV & solar adoption

SYSTEM ARCHITECTURE AND FLOW

1.CT clamp measures real-time current
2.ESP32 processes energy data
3.Data sent to cloud
4.Android app shows live usage & history
5.User takes action to reduce consumption

COMPETITIVE ADVANTAGE

- Real-time energy insights
- Easy retrofit installation
- Designed for Indian homes
- Smart alerts & planning
- No strong real-time competitors

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BUSINESS MODEL OVERVIEW

- One-time hardware purchase
- Optional subscription for analytics
- Value: lower bills & better control
- Scalable and cost-efficient model

Key Partners
- Component & hardware suppliers
- Real estate developers and contractors
- Certified electricians/installers
- Cloud/IoT platforms (e.g., AWS IoT, Blynk)
- Electricity boards, government schemes text

Key Activities
- IoT hardware manufacturing
- AI & software development
- Customer onboarding & support
- Sales & marketing operations
- Partner and utility collaborations of body text

Key Resources
- ESP32-based IoT hardware
- CT clamp energy monitors
- Android app with AI integration
- Skilled dev & support team
- Modular & scalable tech infrastructure

Value Proportions
- Track, control, and optimize electricity use
- Save 15–30% on monthly power bills
- AI-powered usage planner with alerts
- QR-code-based bill explanation
- IoT-based automation to switch off idle devices
- Retrofit-friendly and easy to install
- Surge & fault protection for appliances

Customer Relationships
- Intuitive mobile app interface
- Multichannel support (chat/app/email)
- Installation tutorials and guides
- Feature updates based on user feedback
- Referral and loyalty reward programs

Channels
- Official website & mobile app
- Partner networks (builders, contractors)
- Electronics/smart home retailers
- Collaborations with electricity providers
- Social media, YouTube, local campaigns

Customer Segments
- Urban and semi-urban Indian households
- Small businesses (e.g., clinics, shops)
- Builders and real estate developers
- Electricity boards (integration partners)
- Eco-conscious consumers

Cost Structure
- R&D and prototyping
- Manufacturing and logistics
- Software development and hosting
- Salaries for tech/support team
- Marketing and customer acquisition
- Installation and support services

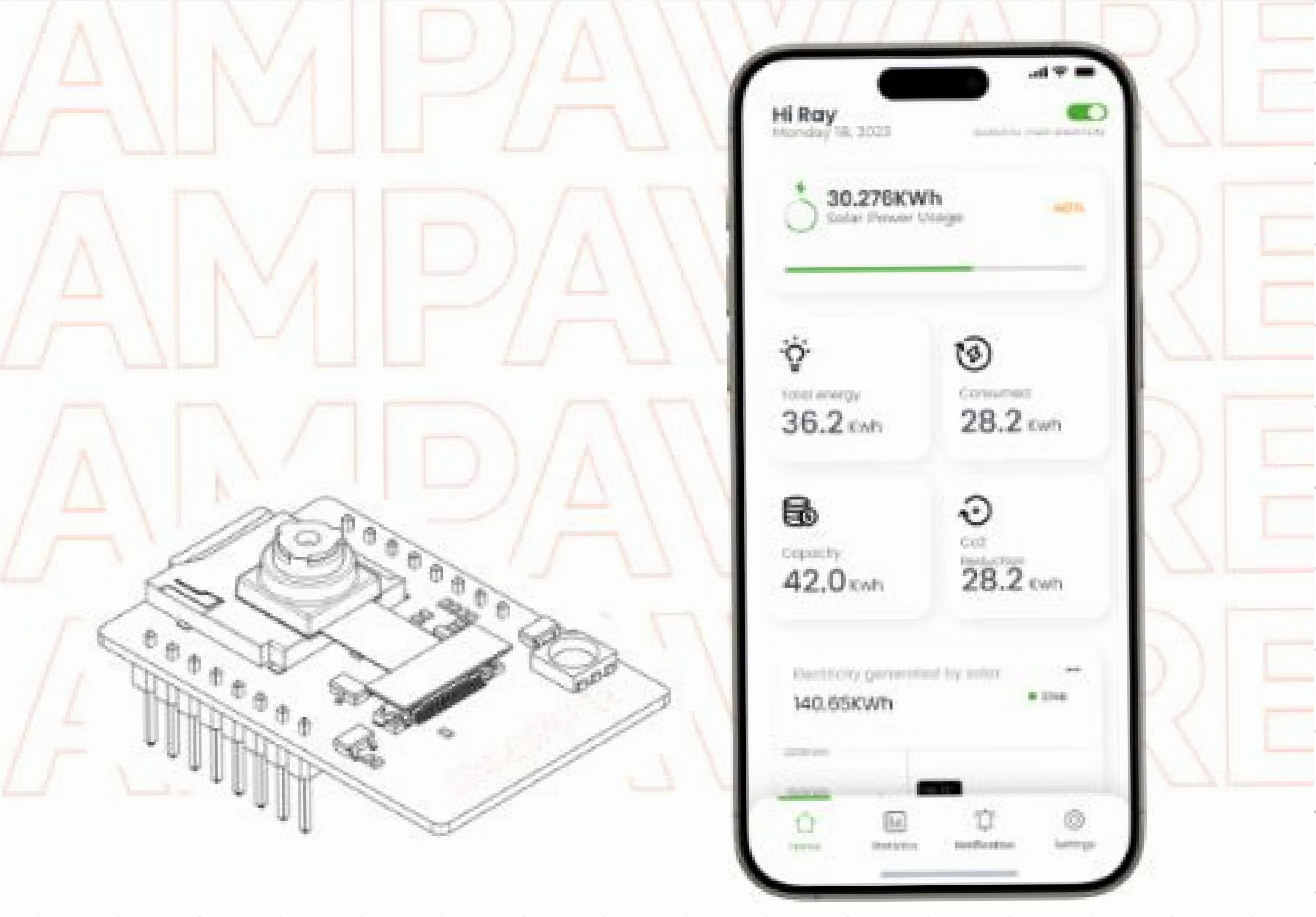
Revenue Streams
- One-time hardware sales (Smart MCBs, switches)
- Monthly/annual subscription (AI planner, automation)
- B2B contracts with builders & electricity boards
- Paid installation & maintenance services

Designed For:	Designed by:	Date:	Version:

Business Model Canvas

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

- ESP32 microcontroller
- CT clamp sensors
- Android application
- Wi-Fi connectivity
- MVP ready in 3 months



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TEAM STRUCTURE AND ROADMAP

AmpSquad Team

Muhammad Mazin KP

Rafan Hatim

Fizza Zainab

Aysha Nabila

3-Month Roadmap

Improve prototype

Pilot testing

Scale deployment

THANK YOU!

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